

JSS2 Basic Science Diagnostic Assessment

(Part 2 – Chemistry)

Curriculum: Cambridge Lower Secondary Stages 7–8 & Nigerian JSS2

Questions 21–40

No.	Question & Options	Ans	Diagnostic Focus
21	The smallest particle of an element that retains its properties is a: A) Molecule B) Atom C) Cell D) Electron	B	Atomic structure.
22	Which particle has a negative charge? A) Proton B) Neutron C) Electron D) Nucleus	C	Subatomic particles.
23	The atomic number of an element is equal to the number of: A) Neutrons B) Protons C) Shells D) Molecules	B	Atomic number concept.
24	Which of these is a chemical change? A) Melting ice B) Rusting iron C) Breaking glass D) Boiling water	B	Physical vs chemical changes.
25	Which method is best for separating sand from water? A) Distillation B) Filtration C) Chromatography D) Evaporation	B	Separation techniques.
26	A substance with pH 2 is: A) Neutral B) Alkaline C) Acidic D) Salty	C	Acids and bases.
27	Which household substance is acidic? A) Soap solution B) Vinegar C) Baking soda D) Detergent	B	Everyday chemistry.
28	What colour does blue litmus paper turn in an acid? A) Blue B) Green C) Red D) Yellow	C	Indicators.
29	Which element is represented by the symbol Na? A) Nitrogen B) Sodium C) Neon D) Nickel	B	Chemical symbols.
30	Which of these is a compound? A) Oxygen B) Iron C) Water D) Gold	C	Elements vs compounds.
31	The process in which a liquid changes to a gas is called: A) Condensation B) Freezing C) Evaporation D) Melting	C	States of matter.
32	Which gas supports burning? A) Nitrogen B) Oxygen C) Carbon dioxide D) Hydrogen	B	Properties of gases.
33	Which instrument measures the mass of a substance? A) Thermometer B) Measuring cylinder C) Beam balance D) Barometer	C	Laboratory apparatus.
34		B	Particle theory.

	Which state of matter has a fixed volume but no fixed shape? A) Solid B) Liquid C) Gas D) Plasma		
35	The periodic table is arranged mainly by increasing: A) Density B) Atomic number C) Mass only D) Size	B	Periodic table.
36	Which of the following is a metal? A) Sulfur B) Oxygen C) Copper D) Chlorine	C	Classification of elements.
37	What is produced when an acid reacts with a base? A) Salt and water B) Oxygen C) Hydrogen only D) Carbon dioxide only	A	Neutralisation.
38	Which safety rule should always be followed in the laboratory? A) Taste chemicals B) Run during experiments C) Wear protective equipment D) Smell chemicals directly	C	Laboratory safety.
39	Which of these is a mixture? A) Air B) Distilled water C) Sodium chloride D) Carbon dioxide	A	Pure substances and mixtures.
40	Diagnostic Insight: Students who struggle with atomic structure, separation techniques and acids/bases require reinforcement before progressing to JSS3 Chemistry.	-	Progress monitoring.